

Linear Algebra Bootcamp 3

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Problem 17. Now we prove the main criterion for diagonalizability. Let $\text{End}_F(W) := \text{Hom}_F(W, W)$ = “square matrices”. Explain why each matrix A determines a map of rings $F[X] \rightarrow \text{End}_F(W)$, sending the poly $p(X)$ to the matrix $p(A)$.

Solution. For any polynomial, $p(x) \in F[X]$, where $p(x) = \sum_{i=0}^k a_i x^i$ we can send it to $p(A) = \sum_{i=0}^k a_i A^i$. This determines a map of rings $F[X] \rightarrow \text{End}_F(W)$. $p(A)$ is also an endomorphism due to being a square matrix since A is also a square matrix. ■

Problem 18. The main theorem is that A is diagonalizable iff we can find a polynomial $p(X)$ which factors completely over F with distinct linear factors. Equivalently, we can find $f_1, \dots, f_k$ distinct elements $f_i \in F$ such that $(A - f_1 I) \dots (A - f_k I) = 0$. Show first that if A is diagonalizable, this is true, and indeed we can take for the f_i the distinct eigenvalues for A .

Solution. Suppose A is diagonalizable. Then by 14, there exists a basis of eigenvectors. By definition of eigenvector, for any eigenvector v there exists an eigenvalue, f such that $Av = fv$ or $(A - fI)v = 0$. By definition, $v \neq 0 \implies A - fI = 0$. Let $f_1, \dots, f_k$ be the distinct eigenvalues. So $\forall i \in [1, \dots, k]$, $A - f_i I = 0$. Thus there are k elements of F such that $(A - f_1 I) \dots (A - f_k I) = 0$. ■

Problem 19. Suppose $p(A) = 0$, and f is an eigenvalue for A . Prove $p(f) = 0$. Informally, whatever poly A satisfies, its eigenvalues satisfy.

Solution. Let f be an eigenvalue of A . Let p be a polynomial such that $p(A) = 0$. By definition of eigenvalue, $\exists v \in V \setminus \{0\} \ni Av = fv$. Let the degree of p be b . $p(A) = \sum_{i=0}^b c_i A^i$. $p(A)v = \left(\sum_{i=0}^b c_i A^i\right)v = \sum_{i=0}^b c_i (A^i v) = \sum_{i=0}^b c_i f^i v = p(f)v = 0 \implies p(f) = 0$. ■

Problem 20. Prove the other direction, that if you have $\dim V$ distinct eigenvalues, then A is diagonalizable.

Solution. Suppose A has $\dim V$ distinct eigenvalues, $f_1, \dots, f_{\dim V}$. Thus by definition of eigenvalue, $\dim V_{f_i} \neq 0$. The addition of $\dim V$ nonzero positive values is at least $\dim V$, so $\dim V_{f_1} + \dots + \dim V_{f_{\dim V}} \geq \dim V$. By 16b, A is diagonalizable and this sum is exactly $\dim V$. ■

Claim 21a. *Prove that $\dim \ker(AB) \leq \dim \ker(A) + \dim \ker(B)$ for any two matrices that have the right size so you can multiply them*

Proof. Let $A : W \rightarrow Z$ and $B : V \rightarrow W$ be linear maps. By problem 10, $\dim V = \dim((AB)_*(V)) + \dim \ker(AB)$, $\dim V = \dim B_*(V) + \dim \ker B$, $\dim W = \dim A_*(W) + \dim \ker A$. This implies, $\dim \ker(AB) = \dim V - \dim((AB)_*(V))$, $\dim \ker B = \dim V - \dim B_*(V)$, $\dim \ker A = \dim W - \dim A_*(W)$.

$\dim \ker B + \dim \ker A - \dim \ker AB = \dim A + \dim((AB)_*(V)) - \dim B_*(V)$. By definition of AB and the image of a linear map, $(AB)_*(V) = \{ABv \mid v \in V\}$ and $A_*(B_*(V)) = \{Aw \mid w \in \{Bv \mid v \in V\}\} = \{ABv \mid v \in V\} = (AB)_*(V)$. Restrict the domain of A to $B_*(V)$ and call this linear map A' . By problem 10, $\dim B_*(V) = \dim \ker A' + \dim(A'_*(B_*(V)))$. $A'_*(B_*(V)) = A_*(B_*(V)) = (AB)_*(V)$. By definition of A' , $\ker A' = B_*(V) \cap \ker A$. So $\dim((AB)_*(V)) = \dim B_*(V) - \dim(\ker A \cap B_*(V))$.

So $\dim \ker B + \dim \ker A - \dim \ker AB = \dim A + \dim((AB)_*(V)) - \dim B_*(V) = \dim A + \dim B_*(V) - \dim(\ker A \cap B_*(V)) - \dim B_*(V) = \dim A - \dim(\ker A \cap B_*(V))$. By definition, $\ker A \cap B_*(V) \subseteq \ker A$, thus $\dim(\ker A \cap B_*(V)) \leq \dim \ker A$. So $\dim \ker B + \dim \ker A - \dim \ker AB = \dim A - \dim(\ker A \cap B_*(V)) \geq 0 \implies \dim \ker B + \dim \ker A - \dim \ker AB \geq 0 \implies \dim \ker B + \dim \ker A \geq \dim \ker AB$. $\square$

Claim 21b. *Show the same for a product a bunch of matrices.*

Proof. ($k = 2$): Base Case proven above in 21a.

Inductive Hypothesis: Suppose for $X_1, \dots, X_k$ where $X_1 \dots X_k$ is a valid matrices ie. multiplication is valid that, $\dim \ker(X_1 \dots X_k) \leq \sum_{i=1}^k \dim \ker X_i$

Inductive Step. Let $X_1, \dots, X_{k+1}$ be $k + 1$ matrices where the multiplication $X_1 \dots X_{k+1}$ is valid. By the base case, $\dim \ker(X_1 \dots X_{k+1}) \leq \dim \ker(X_1 \dots X_k) + \dim \ker X_{k+1}$. By the Inductive Hypothesis, $\dim \ker(X_1 \dots X_k) \leq \sum_{i=1}^k \dim \ker X_i \implies \dim \ker(X_1 \dots X_k) + \dim \ker X_{k+1} \leq \sum_{i=1}^{k+1} \dim \ker X_i \implies \dim \ker(X_1 \dots X_{k+1}) \leq \sum_{i=1}^k \dim \ker X_i$ $\square$

Claim 21c. *A is diagonalizable iff we can find a polynomial $p(X)$ which factors completely over F with distinct linear factors. Equivalently, we can find $f_1, \dots, f_k$ distinct elements $f_i \in F$ such that $(A - f_1 I) \dots (A - f_k I) = 0$.*

Proof. ($\implies$) : Proved in 18. ($\impliedby$) : Suppose you can find a polynomial $p(x)$ which factors completely over F with distinct linear factors, or equivalently, we can find $f_1, \dots, f_k$ distinct elements $f_i \in F$ such that $(A - f_1 I) \dots (A - f_k I) = 0$. Then by 21b, $\dim(\ker((A - f_1 I) \dots (A - f_k I))) = \dim(\ker(0)) = \dim V \leq \sum_{i=1}^k \dim \ker(A - f_i I) = \sum_{i=1}^k \dim V_{f_i}(A)$. Thus by 16b, A is diagonalizable and $\sum_{i=1}^k \dim V_{f_i}(A) = \dim V$. $\square$